Proofs

- **Exercise 1**
  Let $a, b \in \mathbb{R}$ that satisfy $a+b=10$. Prove that $ab \leqslant 25$ or $a^2+b^2<10$.
  - **Solution 1.1**
    Suppose $a, b \in \mathbb{R}$ satisfy $a+b=10$, then $a=10-b$, and we have

$$\begin{aligned} ab &= (10-b)b \\ &= 10b-b^2 \\ &= -(b^2-10b) \\ &= -(b^2-10b+25)+25 \\ &= -(b-5)^2+25 \\ &\leqslant 25 \end{aligned}$$

    So $ab \leqslant 25$, then $ab \leqslant 25$ or $a^2+b^2<10$.
- **Exercise 2**
  Prove that $(\forall x)\, P(x) \vee (\forall x)\, Q(x) \implies (\forall x)\, (P(x) \vee Q(x))$ and refute the converse implication.
  - **Solution 2.1**
    Suppose $(\forall x) P(x) \vee (\forall x) Q(x)$ is true, then $(\forall x) P(x)$ is true or $(\forall x) Q(x)$ is true.
    Suppose $(\forall x) P(x)$ is true, then $\forall x, P(x)$ is true, then $\forall x, P(x) \vee Q(x)$ is true and then $\forall x, (P(x) \vee Q(x))$ is true.
    To check that the converse is false, consider $P(x): x \text{ is even}, Q(x): x \text{ is odd}.$
- **Exercise 3**
  Proof: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
  - **Solution 3.1**
    We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$. Let

$$\begin{aligned} M_1 &:= \min\{a, \max\{b, c\}\} \\ M_2 &:= \max\{\min\{a, b\}, \min\{a, c\}\} \end{aligned}$$

    we want to prove $M_1 \leqslant M_2$ and $M_2 \leqslant M_1$.
    For the case $M_1 \leqslant M_2$, by the definition of $\min$ and $\max$

$$\begin{aligned} & M_1 \leqslant a \wedge M_1 \leqslant \max\{b, c\} \\ \implies & M_1 \leqslant a \wedge (M_1 \leqslant b \vee M_1 \leqslant c) \\ \implies & (M_1 \leqslant a \wedge M_1 \leqslant b) \vee (M_1 \leqslant a \wedge M_1 \leqslant c) \\ \implies & M_1 \leqslant \min\{a, b\} \vee M_1 \leqslant \min\{a, c\} \\ \implies & M_1 \leqslant \max\{\min\{a, b\}, \min\{a, c\}\} \\ \implies & M_1 \leqslant M_2 \end{aligned}$$

    For the case $M_2 \leqslant M_1$, by the definition of $\min$ and $\max$, we have

$$\min\{a, b\} \leqslant a, \min\{a, b\} \leqslant b \leqslant \max\{b, c\}$$

    then $\min\{a, b\} \leqslant \min\{a, \max\{b, c\}\}$.
    The other inequality is almost the same. Then $M_2 = \max\{\min\{a, b\}, \min\{a, c\}\}$.
  - **Solution 3.2**
    We want to prove $\forall a, b, c \in \mathbb{R}, \min\{a, \max\{b, c\}\} = \max\{\min\{a, b\}, \min\{a, c\}\}$.
    Suppose $a, b, c \in \{0, 1\}$. We split into two cases depending on the value of a.
    Case 1 $a=0$

$$\begin{aligned} \text{LHS} &= \min\{0, \max\{b, c\}\} = 0 \\ \text{RHS} &= \max\{\min\{0, b\}, \min\{0, c\}\} = \max\{0, 0\} = 0 \end{aligned}$$

Thus $\text{LHS} = \text{RHS}$.
Case 2 $a = 1$

$$\begin{gathered}\text{LHS} = \min\{1, \max\{b, c\}\} = \max\{b, c\} \\ \text{RHS} = \max\{\min\{1, b\}, \min\{1, c\}\} = \min\{b, c\}\end{gathered}$$

Thus $\text{LHS} = \text{RHS}$.
Since equality holds for $a = 0$ and $a = 1$, the identity is true for all $a, b, c \in \{0, 1\}$.

- **Exercise 4**
  Let $a_1, a_2, \cdots, a_7 \in \mathbb{Z}$, prove that among these seven integers, one can always choose four numbers whose sum is even.
  - **Solution 3.1**
    Suppose $a_1, a_2, \cdots, a_7 \in \mathbb{Z}$. Let K be the number of odd integers among $a_1, a_2, \cdots, a_7$.
    If $4 \leqslant K \leqslant 7$, then take 4 of these odd integers, their sum is even:
    $$(2a+1) + (2b+1) + (2c+1) + (2d+1) = 2(a+b+c+d+2)$$
    If $0 \leqslant K < 4$, then take 4 of these even integers, their sum is even:
    $$2a + 2b + 2c + 2d = 2(a+b+c+d)$$
    In both cases, we obtain that we can always choose four numbers whose sum is even.
- **Exercise 5**
  Prove that $\forall n \in \mathbb{Z}$, if n is odd, then $3n$ is odd.
  - **Solution 5.1**
    Suppose $n \in \mathbb{Z}$, and n is odd. Then, there exists an integer $k \in \mathbb{Z}$ such that $n = 2k + 1$. Then
    $$\begin{aligned} 3n &= 3(2k+1) \\ &= 6k + 3 \\ &= 2(3k+1) + 1 \end{aligned}$$
    Since $3k + 1$ is an integer, $3n$ is odd.
- **Exercise 6**
  For every $a, b, c \in \mathbb{Z}$, prove that if $a \mid b$ and $b \mid c$, then $a \mid c$.
  - **Solution 6.1**
    Suppose $a, b, c \in \mathbb{Z}$ and $a \mid b \wedge b \mid c$. Then there exist $k, q \in \mathbb{Z}$ such that $b = ka$ and $c = qb$. Then $c = qka$, and since $qk \in \mathbb{Z}$, by definition, $a \mid c$.
- **Exercise 7**
  Let $n \in \mathbb{Z}$. Prove that if n^2 is even, then n is even.
  - **Solution 7.1**
    We will prove the contrapositive: $\forall n \in \mathbb{Z}$, if n is odd, then n^2 is odd.
    Suppose $n \in \mathbb{Z}$ is odd. Then $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$. Then
    $$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$
    is odd because $2k^2 + 2k$ is an integer.
- **Exercise 8**
  Prove that there is no smallest positive rational number.
  - **Solution 4.7.1**
    By contradiction, suppose there exists a smallest positive rational number r. Then $\frac{r}{2}$ is a smaller positive rational number. This is a contradiction because r was assumed to be the smallest. Thus, there doesn't exist a smallest positive rational number.

- **Exercise 9**

  $\forall \epsilon > 0$, prove that $\exists N \in \mathbb{N}$ such that $\forall n \geqslant N$, $\dfrac{1}{n} < \epsilon$.

  - **Solution 9.1**

    Suppose $\epsilon > 0$. We will first prove that $\exists N \in \mathbb{N}$ such that $\dfrac{1}{N} < \epsilon$.
    The Archimedean Property states that for any real number $x \in \mathbb{R}$, there exists a natural number $N \in \mathbb{N}$ such that $N > x$. By setting $x = \dfrac{1}{\epsilon}$ in the Archimedean Property, we guarantee that $\exists N \in \mathbb{N}$ satisfying $N > \dfrac{1}{\epsilon}$, and then $\epsilon > \dfrac{1}{N}$. Then if $n \geqslant N$, $\dfrac{1}{N} \geqslant \dfrac{1}{n}$ and $\dfrac{1}{n} < \epsilon$.